

Mehrdad Mohammadi

CONTACT	E-mail: mehrdad3@illinois.edu	Website: mehrdadmhmdi.github.io
CURRENT POSITION	Postdoctoral Research Fellow, <i>Beckman Institute</i> <i>The Science and Technology Center for Quantitative Cell Biology, U of Illinois Urbana-Champaign</i>	September 2026 - Current
WORK EXPERIENCE	Data Scientist and Machine Learning Intern, Bayer AG Research Fellow, Bioacoustics Research Laboratory (<i>Department of Electrical and Computer Engineering</i>)	May 2023 - December 2023 May 2022-May 2023
EDUCATION	University of Illinois Urbana-Champaign, USA Ph.D. Statistics	2026
	University of Illinois Urbana-Champaign, USA M.S. Mathematics	2023
	University of Illinois Urbana-Champaign, USA M.S. Statistics	2023
	Pennsylvania State University, USA M.A. Economics	2020
	University of Bologna, Italy M.S. Economics, LMEC	2018
	Petroleum University of Technology, Iran B.S. Finance and Accounting	2015
IN-REVIEW WORKS	Mohammadi, M., Zheng, Q., & Zhu, R. (2026). “Vector-Valued Distributional Reinforcement Learning Policy Evaluation: A Hilbert Space Embedding Approach”. arxiv.org/abs/2601.18952 Mohammadi, M., “Toscani-Fourier Distance on Probability Measures: Wasserstein Control, Topological Equivalence on Model Classes, and Duality” Mohammadi, M., Simpson, D. G. “SILK: Shift-invariant Landmark Kernels For Dynamic Risk Prediction With An Unknown Time Origin” Mohammadi, M., Simpson, D. G. , Villegas-Downs, M., McFarlin, B. L., Han, A., & O’Brien Jr, W. D. “Heterogeneous Phase Shifts in Gestational Trajectories: A Quantile Regression Approach to Spontaneous Preterm Birth”	
PUBLICATIONS	Villegas-Downs M.*, Mohammadi, M.*, Han A, O’Brien WD Jr, Schlaeger JM, McFarlin BL. “Characterization of Cervical Remodeling During Pregnancy and the Postpartum Period: A Case Series Analysis”. <i>American Journal of Case Reports</i>, (2026)	
(GOOGLE SCHOLAR)	Villegas-Downs M.*, Mohammadi, M.*, Han, A., O’Brien Jr, W. D., Simpson, D. G., Peters, T. A., Schlaeger,J.M. , and & McFarlin, B. L. “Trajectory of Postpartum Cervical Remodeling in Women Delivering Full-term and Spontaneous Preterm: Sensitivity to Quantitative Ultrasound Biomarkers”, <i>Ultrasound in Medicine & Biology</i>,(2024)	
* EQUAL CONTRIBUTION	McFarlin, B. L.*, Villegas-Downs, M.*, Mohammadi, M.*, Han, A., Simpson, D. G., & O’Brien Jr, W. D. (2023). “Enhanced identification of women at risk for preterm birth via quantitative ultrasound: a prospective cohort study.” <i>American Journal of Obstetrics & Gynecology MFM</i>, 101250. McFarlin, B. L., Liu, Y., Villegas-Downs, M., Mohammadi, M., Simpson, D. G., Han, A., & O’Brien Jr, W. D. (2023). “Predicting Spontaneous Pre-term Birth Risk Is Improved When Quantitative Ultrasound Data Are Included With Historical Clinical Data.” <i>Ultrasound in Medicine & Biology</i>, 49(5), 1145-1152.	
REVIEW SERVICES	Journal of Computational and Graphical Statistics Journal of Statistical Computation and Simulation	

TALKS	2026: The Joint Statistical Meetings (Boston, MA) 2025: The Joint Statistical Meetings (Nashville, TN), Bohrer Workshop (UIUC)	
AWARDS AND HONORS	Larine Y. Cowan Make A Difference Leadership Award Illinois International Graduate Achievement Award Social Justice Bridge Builder Graduate Award University of Illinois Teachers Ranked as Excellent by Students The Acoustical Society of America Conference, Poster Presentation Award University of Illinois, Department of Statistics Leadership Award University of Illinois, Block Grant Ph.D. Fellowship University of Bologna, Best Student University-Wide Award	2025 2025 2025 2024 2023 2023 2020-2021 2017
SERVICES AND ACTIVITIES	Larine Y. Cowan Award Review Committee Advisor to the University of Illinois Chancellor on Student Affairs Inaugural Director Hiring Committee Member, <i>SALAAM MENA Cultural Center at UIUC</i> Founding Committee Member, <i>SALAAM MENA Cultural Center at UIUC</i> Founder and President, <i>Statistics Doctoral Student Association at UIUC</i> Treasurer and Director, <i>Iranian Music Heritage at UIUC</i> Treasurer and Director , <i>Iranian Cultural Association at UIUC</i> Basketball Referee, <i>Iran Basketball Federation</i> Open Water Diver (CMAS certificate), <i>Italian Underwater Activities Federation</i>	
SOFTWARE	Python: (KE-DRL) <i>Kernel Embedding Distributional Reinforcement Learning</i>	
LANGUAGES	Persian(Native), Kurdish (Native), English (Fluent), Italian (Elementary)	
PROGRAMMING & SCIENTIFIC COMPUTING	Python (PyTorch), R, SQL - Linux & HPC (SLURM, GPU computing) - Git, L ^A T _E X, MS Office.	
REFERENCES	Dr. Ruoqing Zhu Associate Professor Department of Statistics University of Illinois Urbana-Champaign email: rqzhu@illinois.edu	Dr. Douglas G. Simpson Professor Department of Statistics University of Illinois Urbana-Champaign email: dgs@illinois.edu
	Dr. Qi Zheng Associate Professor Department of Bioinformatics and Biostatistics University of Louisville email: qi.zheng@louisville.edu	Dr. Jingbo Liu Assistant Professor Department of Statistics University of Illinois Urbana-Champaign email: jingbol@illinois.edu